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Inventor(s)	Mao; Zhengda et al.

Armrest assembly with internal motion damper

Abstract

An armrest assembly includes an armrest body that defines an interior space. A first shaft extends from a first side of the armrest body along an armrest axis and is attached to the armrest body for relative rotational movement about the armrest axis. A second shaft extends from a second side of the armrest body along the armrest axis and is attached to the armrest body for relative rotational movement about the armrest axis. A rotational damper is located in the interior space of the armrest body and includes a first damper part and a second damper part attached to the first damper part for relative rotational movement. The first damper part is fixed to the first shaft, and the second damper part is fixed to the armrest body.

Inventors: Mao; Zhengda (Shanghai, CN), Zhao; Mingzhe (Shanghai, CN), Wu; Yili (Shanghai, CN)

Applicant: Lear Corporation (Southfield, MI)

Family ID: 88974886

Assignee: Lear Corporation (Southfield, MI)

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4018104	12/1976	Bland	74/513	G05G 5/22
4491436	12/1984	Easton	403/131	F16C 11/0614
4586750	12/1985	Vogel	297/411.32	A47C 7/543
5076645	12/1990	Yokota	297/411.32	B60N 2/753
5342115	12/1993	De Filippo	403/104	B60N 2/763
5395160	12/1994	Krebs	297/411.3	A47C 7/543
5937482	12/1998	Horng	16/337	G06F 1/1681
6578922	12/2002	Khedira	297/411.3	B60N 2/753
6671928	12/2003	Huang	16/337	F16M 11/10
7077477	12/2005	Syrowik	297/411.3	B60N 2/753
7140688	12/2005	Hann	297/411.32	B60N 2/753
8516668	12/2012	Weber	297/411.32	B60N 2/78
8960786	12/2014	Henke	297/14	B60N 2/757
9132755	12/2014	Rosenbrock	N/A	B60N 2/753
9162597	12/2014	Charles et al.	N/A	N/A
9227538	12/2015	Girbinger et al.	N/A	N/A
9776538	12/2016	Cheng	N/A	B60N 2/682
9950651	12/2017	Delling	N/A	B60N 2/933
10131258	12/2017	Akai	N/A	B60N 2/682
10308152	12/2018	Bohlke	N/A	B60N 2/753
10336229	12/2018	Johnson	N/A	B60N 2/757
10434912	12/2018	Zapf	N/A	N/A
10457177	12/2018	Nuss et al.	N/A	N/A
10562424	12/2019	Stuiber et al.	N/A	N/A
10710483	12/2019	Keller	N/A	N/A
10889218	12/2020	Vetere, II	N/A	N/A
10994640	12/2020	Powell	N/A	B60N 2/793
11351899	12/2021	Kobayashi	N/A	B60N 2/75
11752910	12/2022	Boddenberg	297/411.32	B60N 2/767
2018/0339627	12/2017	Keller et al.	N/A	N/A
2020/0345146	12/2019	Himmelhuber	N/A	B60N 2/777
2022/0203900	12/2021	Nishio	N/A	E05F 3/14

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102017214628	12/2017	DE	N/A
102017121229	12/2018	DE	N/A
102020120824	12/2020	DE	N/A
3572275	12/2018	EP	N/A
2882306	12/2005	FR	N/A
3052410	12/2016	FR	N/A
3070919	12/2018	FR	N/A
3073464	12/2018	FR	N/A
3074111	12/2018	FR	N/A
3080584	12/2018	FR	N/A
3082467	12/2018	FR	N/A
3096620	12/2019	FR	N/A
2021009169	12/2020	WO	N/A

Primary Examiner: Islam; Syed A

Attorney, Agent or Firm: MacMillan, Sobanski & Todd, LLC

Background/Summary

BACKGROUND OF THE INVENTION

(1) This invention relates in general to a seat that includes an armrest assembly for comfortably supporting an arm of an occupant of the seat. In particular, this invention relates to an improved structure for such an armrest assembly that is movable between usage and storage positions.

(2) Vehicles, such as passenger cars, typically include one or more seats for supporting occupants therein. In many instances, a vehicle seat includes an armrest assembly that is provided for comfortably supporting an arm of an occupant of the seat. In many instances, the armrest assembly is pivotably movable relative to the seat between a lowered usage position and a raised storage position. When located in the lowered usage position, the armrest assembly extends generally horizontally adjacent to a bottom portion of the seat and is available for use by the occupant therein. When located in the raised storage position, the armrest assembly extends generally vertically adjacent to a back portion of the seat. One example of such an armrest assembly is disclosed in U.S. Pat. No. 10,336,229, the disclosure of which is incorporated herein by reference.

(3) It would be desirable to provide an improved structure for an armrest assembly for a seat that is movable between usage and storage positions.

SUMMARY OF THE INVENTION

(4) This invention relates to improved structure for an armrest assembly for a seat that is movable between usage and storage positions. The armrest assembly includes an armrest body that defines an interior space. A first shaft extends from a first side of the armrest body along an armrest axis and is attached to the armrest body for relative rotational movement about the armrest axis. A second shaft extends from a second side of the armrest body along the armrest axis and is attached to the armrest body for relative rotational movement about the armrest axis. A rotational damper is located in the interior space of the armrest body and includes a first damper part and a second damper part attached to the first damper part for relative rotational movement. The first damper part is fixed to the first shaft, and the second damper part is fixed to the armrest body.

(5) Various aspects of this invention will become apparent to those skilled in the art from the

following detailed description of the preferred embodiment, when read in light of the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side elevational view of a seat that includes an armrest assembly in accordance with this invention, wherein the armrest assembly is shown in a stowed position relative to the seat.
- (2) FIG. 2 is a side elevational view similar to FIG. 1 showing the armrest assembly in a usage position relative to the seat.
- (3) FIG. 3 is an enlarged perspective view of a portion of the armrest assembly shown in FIG. 1.
- (4) FIG. 4 is a cross-sectional view taken along line 4-4 of FIG. 3.
- (5) FIG. 5 is an exploded perspective view of a first shaft assembly of the armrest assembly shown in FIGS. 3 and 4.
- (6) FIG. 6 is a cross-sectional view of the first shaft assembly shown partially assembled.
- (7) FIG. 7 is an exploded perspective view of a second shaft assembly of the armrest assembly shown in FIGS. 3 and 4.
- (8) FIG. 8 is a cross-sectional view of the second shaft assembly shown partially assembled.
- (9) FIG. 9 is a cross-sectional view similar to FIG. 8 showing the second shaft assembly in an inserted position relative to the seat.
- (10) FIG. 10 is a cross-sectional view similar to FIG. 9 showing the second shaft assembly in an installed position relative to the seat.
- (11) FIG. 11 is a cross-sectional view taken along line 11-11 of FIG. 4.
- (12) FIG. 12 is a cross-sectional view taken along line 12-12 of FIG. 4.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

- (13) Referring now to the drawings, there is illustrated in FIG. 1 a side elevational view of a seat, indicated generally at 10, in accordance with this invention. The illustrated seat 10 is a conventional seat for a vehicle that includes a seat bottom portion 12 and a seat back portion 13, but the seat 10 may be any desired type of seat. The seat 10 also includes an armrest assembly, indicated generally at 14. The armrest assembly 14 includes an armrest 16 that is movable relative to the seat 10 between a stowed position (illustrated in FIG. 1) and a usage position (illustrated in FIG. 2). To accomplish this, the armrest 16 may be rotated relative to the seat 12 about an armrest axis 18. When the armrest 16 is in the usage position, a trim surface 20 is presented for use by an occupant of the seat 12.
- (14) Referring to FIG. 3, there is illustrated a perspective view of part of the armrest assembly 14. The trim surface 20 of the armrest 16 is not shown in FIG. 3 so that interior parts of the armrest assembly 14 are visible. The armrest assembly 14 includes an armrest body 22. The illustrated armrest body 22 is molded from plastic, but may be made from any desired material using any desired process. The armrest body 22 includes a base 24 and two walls 26 that define an interior space, indicated generally at 28.
- (15) The armrest assembly 14 includes a first shaft assembly, indicated generally at 30 that is located on a first side 32 of the armrest body 22. The armrest assembly 14 also includes a second shaft assembly, indicated generally at 34, that is located on a second side 36 of the armrest body 22.
- (16) Referring to FIG. 4, there is illustrated a cross-sectional view taken along line 4-4 of FIG. 3. The cross-section illustrated in FIG. 4 is taken along the armrest axis 18. The first shaft assembly 30 includes a first shaft 38. The first shaft 38 extends from the first side 32 of the armrest body 22 along the armrest axis 18. The first shaft 38 engages a first bracket 40 that is attached to the seat assembly 10. The second shaft assembly 34 includes a second shaft 42. The second shaft 42 extends from the second side 36 of the armrest body 22 along the armrest axis 18. The second shaft

42 engages a second bracket **44** that is also attached to the seat assembly **10**.

(17) Referring now to FIG. 5, there is illustrated an exploded perspective view of the first shaft assembly **30**. Referring to FIG. 6, there is shown an enlarged view of a cross-sectional view similar to FIG. 4, with the first shaft assembly **30** shown partially assembled. The first shaft **38** is made of metal, but may be made of any desired material. The first shaft **38** extends from a first keyed end **46** to a first inner end **48** and includes a bushing mount **50** therebetween. When the armrest assembly **14** is assembled, the first shaft **38** extends along the armrest axis **18**, as shown in FIG. 6. The first shaft **38** includes a rotation support portion **52** between the keyed end **46** and the bushing mount **50**. The illustrated rotation support portion **52** has a circular cross-sectional shape around the armrest axis **18**, but may have any desired shape. The first shaft **38** also includes a fixed support portion **54** between the bushing mount **50** and the first inner end **48**. The illustrated fixed support portion **54** has a hexagonal cross-sectional shape around the armrest axis **18**, but may have any desired shape.

(18) The first shaft assembly **30** includes a first insert **56**. The first insert **56** is made of metal but may be made of any desired material. As shown in FIG. 6, when the armrest assembly **14** is assembled, the first insert **56** is fixed on the armrest body **32**. The first insert **56** includes an insert opening **58** that extends through the first insert **56**, and the armrest axis **18** passes through the insert opening **58**. The insert opening **58** has a circular cross-sectional shape around the armrest axis **18**.

(19) The first shaft assembly **30** includes a bushing **60**. The bushing **60** is made of plastic, but may be made of any desired material. The bushing **60** is located around the rotation support portion **52** of the first shaft **38**, and the bushing **60** engages the bushing mount **50**. As shown in FIG. 6, the bushing **60** is fixed to the first shaft **38** and is located in the insert opening **58** between the first shaft **38** and the first insert **56**. The bushing **60** provides a relatively low friction contact with the first insert **56** to facilitate rotation of the armrest body **22** relative to the first shaft **38**.

(20) The first shaft assembly **30** includes a friction washer **62** and a retainer **64** that are located between the first insert **56** and the keyed end **46** of the first shaft **38**. The retainer **64** engages a slot **66** in the surface of the rotation support portion **52** and presses the friction washer **62** into engagement with the first insert **56**. As shown in FIG. 6, when the first shaft assembly **30** is assembled, the first insert **56** is captured between the bushing **60** and the friction washer **62**. Thus, the first shaft assembly **30** is prevented from moving relative to the armrest body **22** in a direction parallel to the armrest axis **18**. However, the first shaft **38** and the bushing **60** are able to rotate relative to the armrest body **22** about the armrest axis **18**.

(21) The first shaft assembly **30** also includes a rotational damper **68**. The rotational damper **68** includes a damper inner **70** and a damper shell **72**. The illustrated damper inner **70** is a first damper part and the illustrated damper shell **72** is a second damper part. The damper shell **72** is attached to the damper inner **70** for relative rotation about the armrest axis **18**. The illustrated rotational damper **68** is an oil damper that includes a viscous fluid (not shown) to limit the rate of movement of the damper shell **72** relative to the damper inner **70**, as is known in the art. The rotational damper **68** is located on the fixed support portion **54** of the first shaft **38**, and the damper inner **70** is fixed to the first shaft **38**. A retaining washer **74** engages a slot **76** in the fixed support portion **54** of the first shaft **38** and is fixed to the first shaft **38** between the rotational damper **68** and the inner end **48**. As shown in FIG. 6, when the first shaft assembly **30** is assembled, the rotational damper **68** is captured between the retaining washer **74** and the bushing mount **50** to prevent the rotational damper **68** from moving relative to the first shaft **38** in a direction parallel to the armrest axis **18**. Additionally, when the first armrest assembly **14** is assembled, the rotational damper **68** is located in the interior space **28** of the armrest body **22**. The operation of the rotational damper **68** will be described below.

(22) Referring now to FIG. 7, there is illustrated an exploded perspective view of the second shaft assembly **34**. Referring to FIG. 8, there is shown an enlarged view of a cross-sectional view similar to FIG. 4, with the second shaft assembly **34** shown partially assembled. The second shaft **42** is

made of metal, but may be made of any desired material. The second shaft **42** extends from a second keyed end **78** to a second inner end **80** and includes a bushing mount **82** therebetween. When the armrest assembly **14** is assembled, the second shaft **42** extends along the armrest axis **18**, as shown in FIG. **8**. The second shaft **42** includes a rotation support portion **84** between the second keyed end **78** and the bushing mount **82**. The illustrated rotation support portion **84** has a circular cross-sectional shape around the armrest axis **18**, but may have any desired shape. In the illustrated embodiment, the bushing mount **82** is located adjacent to the second inner end but may be in any desired location on the second shaft **42**.

(23) The second shaft assembly **34** includes a second insert **86**. The second insert **86** is made of metal but may be made of any desired material. As shown in FIG. **8**, when the armrest assembly **14** is assembled, the second insert **86** is fixed on the armrest body **32**. The second insert **86** includes an insert opening **88** that extends through the second insert **86**, and the armrest axis **18** passes through the insert opening **88**. The insert opening **88** has a circular cross-sectional shape around the armrest axis **18**.

(24) The second shaft assembly **34** includes a bushing **90**. The bushing **90** is made of plastic but may be made of any desired material. The bushing **90** is located around the rotation support portion **84** of the second shaft **42**, and the bushing **90** engages the bushing mount **82**. As shown in FIG. **8**, the bushing **90** is fixed to the second shaft **42** and is located in the insert opening **88** between the second shaft **42** and the second insert **86**. The bushing **90** provides a relatively low friction contact with the second insert **86** to facilitate rotation of the armrest body **22** relative to the second shaft **42**.

(25) The second shaft assembly **34** includes a resilient washer **92** that is located between two friction washers **94** and **96** that are provided between the second insert **86** and the second keyed end **78** of the second shaft **42**. A retainer **98** engages a slot **100** in the surface of the rotation support portion **52** and presses the friction washer **96** into engagement with the resilient washer **92**. The resilient washer **92** is further pressed into engagement with the friction washer **94**. As shown in FIG. **8**, when the second shaft assembly **34** is assembled, the second insert **86** is captured between the bushing **90** and the friction washer **94** and thus prevents the second shaft assembly **34** from moving relative to the armrest body **22** in a direction parallel to the armrest axis **18**. However, the second shaft **42** and the bushing **90** are able to rotate relative to the armrest body **22** about the armrest axis **18**.

(26) The assembled armrest assembly **14**, including the first shaft assembly **30** and the second shaft assembly **34**, is adapted to be connected to the seat **12** as described below. Referring to FIG. **6**, the first keyed end **46** of the first shaft **38** engages the first bracket **40**. As best shown in FIG. **5**, the first keyed end **46** has a non-circular shape and engages the first bracket **40** to prevent rotation of the first shaft **38** relative to the first bracket **40**. Referring to FIG. **9**, there is illustrated a cross-sectional view of the second shaft assembly **34** similar to FIG. **8**. As shown in FIG. **9**, the second keyed end **78** of the second shaft **42** is moved relative to the armrest body **22** in an insertion direction **102** parallel to the armrest axis **18** toward the first shaft assembly **30** (not shown in FIG. **9**). As previously described, the bushing **90** and the retainer **98** are fixed to the second shaft **42**. Thus, the bushing **90** and the retainer **98** are also moved in the insertion direction **102** relative to the armrest body **22** along with the second shaft **42**. The retainer **98** additionally causes the friction washer **96** to move in the insertion direction **102** relative to the friction washer **94**. As a result, the resilient washer **92** is compressed between the friction washer **94** and the friction washer **96**. The second shaft assembly **34** is then located in an inserted position shown in FIG. **9**.

(27) While the second shaft assembly **34** is in the inserted position, the second keyed end **78** of the second shaft **42** is then positioned adjacent to the second bracket **44**. Because the second keyed end **78** has been moved in the insertion direction **102**, the second shaft **42** can be positioned between the first bracket **40** and the second bracket **44** when the first shaft is engaged with the first bracket **40**. The second keyed end **78** is then moved opposite the insertion direction **102** so that the second keyed end **78** engages the second bracket **44**. This locates the second shaft assembly **34** in an

installed position shown in FIG. 10. As previously described, when the second shaft assembly 34 is in the inserted position shown in FIG. 9, the resilient washer 92 is compressed. This can be done, for example, by an operator applying force to the second shaft 42 in the insertion direction 102. The resilient washer 92 applies a force that biases the second shaft 42 opposite the insertion direction 102. When the force on the second shaft 42 is removed, the second shaft 42 is moved from the inserted position to the installed position shown in FIG. 10. As best shown in FIG. 7, the second keyed end 78 has a non-circular shape and engages the second bracket 44 to prevent rotation of the second shaft 42 relative to the second bracket 44.

(28) Referring back to FIG. 5, the first shaft assembly 30 also includes a screw 104. As shown in FIG. 4, the screw 104 is attached to the first keyed end 46 of the first shaft 38 to fix the first shaft 38 in position relative to the first bracket 40. Similarly, referring to FIG. 7, the second shaft assembly 34 includes a screw 106. As shown in FIG. 4, the screw 106 is attached to the second keyed end 78 of the second shaft 42 to fix the second shaft 42 in place relative to the second bracket 44.

(29) When the armrest assembly 14 is attached to the seat 12, the first shaft 38 and the second shaft 42 are fixed relative to the seat 12, and the armrest body 22 is adapted to be rotated relative to both the first shaft 38 and the second shaft 42 about the armrest axis 18. This allows the armrest 16 to be rotated relative to the seat 12. Referring to FIG. 11, there is illustrated an end view of the first shaft assembly 30 taken along line 11-11 of FIG. 4. The rotational damper 68 includes a grip 107 on the damper shell 72 that engages the armrest body 22 to fix the damper shell 72 relative to the armrest body 22. As previously described, the damper inner 70 is fixed relative to the first shaft 38. Thus, when the armrest 16 is moved relative to the seat 12 between the stowed position shown in FIG. 1 and the usage position shown in FIG. 2, the rotational damper 68 is adapted to dampen the movement of the armrest 16 relative to the seat 12.

(30) Referring to FIG. 2, the illustrated armrest 16 is adapted to be moved relative to the seat 12 from the stowed position shown in FIG. 1 through a first range of motion 108. During the first range of motion 108, the movement of the armrest 16 relative to the seat 12 is not restricted by the rotational damper 68. The illustrated armrest 16 is further adapted to be moved relative to the seat 12 through a second range of motion 110 to the usage position. During the second range of motion 110, the movement of the armrest 16 relative to the seat 12 is restricted by the rotational damper 68. Thus, the rotational damper 68 prevents the armrest 16 from suddenly moving to the usage position. In the illustrated embodiment, both the first range of motion 108 and the second range of motion 110 are approximately fifty-six degrees of rotation. However, the first range of motion 108 and the second range of motion 110 may be any desired size or sizes.

(31) Referring to FIG. 12, there is illustrated a cross-sectional view taken along line 12-12 of FIG. 4. The view illustrated in FIG. 12 is taken through the first shaft assembly 30 and parallel to FIG. 11. The armrest assembly 14 includes a detent assembly, indicated generally at 112. The detent assembly 112 is adapted to retain the armrest 16 in a position relative to the seat 12. The detent assembly 112 includes a stop 114 and a detent body 116. The illustrated stop 114 is made of resilient rubber, but may be made of any desired material. The stop 114 is fixed on the armrest body 22 and is located in the interior space 28 of the armrest body 22. The detent body 116 is made from molded plastic, but may be made of any desired material using any desired process. The illustrated detent body 116 is molded as part of the bushing 60, but may be a separate part of the armrest assembly 14. The detent body 116 is fixed relative to the first shaft 38 and is located in the interior space 28 of the armrest body 22. Because the first shaft 38 is fixed relative to the seat 12, when the armrest 16 is moved relative to the seat 12, the stop 114 is moved relative to the detent body 116.

(32) When the armrest 16 is in the usage position, the stop 114 is located in a first detent 118 on the detent body 116, as shown in FIG. 12. The first detent 118 is defined, at least in part, by a first rise 120. When the armrest 16 is moved from the usage position toward the stowed position, the stop 114 engages the first rise 120 to resist movement of the armrest 16 relative to the seat 12. Further,

when the armrest **16** is in the stowed position, the stop **114** is located in a second detent **122** on the detent body **116**. The second detent **122** is defined, at least in part, in part by a second rise **124**. When the armrest **16** is moved from the stowed position toward the usage position, the stop **114** engages the second rise **124** to resist movement of the armrest **16** relative to the seat **12**.

(33) The detent body **116** includes a detent surface **126** that is located between the first rise **120** and the second rise **124**. The detent surface **126** is a curved surface that extends from the first rise **120** to the second rise **124**. The illustrated detent surface **126** is nearest the armrest axis **18** adjacent to the second rise **124** and moves away from the armrest axis **18** at a substantially constant slope relative to the first rise **120**. The detent surface **126** is located a first distance **128** from the armrest axis **18** adjacent to the first rise **120** and a second distance **130** from the armrest axis **18** adjacent to the second rise **124**. In the illustrated embodiment, the first distance **128** is greater than the second distance **130**. However, the detent surface **126** may have any desired shape. When the armrest **16** is moved from the stowed position to the usage position through the first range of motion **108** and the second range of motion **110**, the stop **114** engages the detent surface **126**.

(34) The detent body **116** includes a first space **132** that is located between the first rise **120** and the armrest axis **18**. The first space **132** is an open space that extends through the detent body **116** in a direction that is parallel to the armrest axis **18**. The first space **132** extends between the first detent **118** and the armrest axis **18**, and also between a portion of the detent surface **126** and the armrest axis **18**. The first space **132** provides an amount of flexibility, allowing at least a portion of the detent body **116** to be deformed when the stop **114** engages the first rise **120**. The first space **132** includes an arcuately-shaped middle portion **132a** that extends between two end portions **132b** located at opposed ends of the middle portion **132a**. The illustrated end portions **132b** extend closer to the armrest axis **18** than the middle portion **132a**. However, the first space **132** may have any desired shape.

(35) The detent body **116** also includes a second space **134** that is located between the second rise **124** and the armrest axis **18**. The second space **134** is also an open space that extends through the detent body **116** in a direction that is parallel to the armrest axis **18**. The second space **134** extends between the second detent **122** and the armrest axis **18**, and also between a portion of the detent surface **126** and the armrest axis **18**. The second space **130** provides an amount of flexibility, allowing at least a portion of the detent body **116** to be deformed when the stop **114** engages the second rise **124**. The second space **134** includes an arcuate middle portion **134a** that extends between two end portions **134b** located at opposed ends of the middle portion **134a**. The illustrated end portions **134b** extend closer to the armrest axis **18** than the middle portion **134a**. However, the second space **134** may have any desired shape.

(36) The principle and mode of operation of this invention have been explained and illustrated in its preferred embodiment. However, it must be understood that this invention may be practiced otherwise than as specifically explained and illustrated without departing from its spirit or scope.

Claims

1. An armrest assembly comprising: an armrest body that defines an interior space; a first shaft that extends from a first side of the armrest body along an armrest axis and is attached to the armrest body for relative rotational movement about the armrest axis; a second shaft that extends from a second side of the armrest body along the armrest axis and is attached to the armrest body for relative rotational movement about the armrest axis; and a rotational damper located in the interior space of the armrest body, the rotational damper including a first damper part and a second damper part attached to the first damper part for relative rotational movement, wherein the first damper part is fixed to the first shaft, and the second damper part is fixed to the armrest body.
2. The armrest assembly of claim 1, wherein the first shaft is attached to the armrest body for relative rotational movement through a first range of motion and a second range of motion, and

wherein the rotational damper is adapted to limit movement of the first shaft relative to the armrest body through the second range of motion.

3. The armrest assembly of claim 1, wherein the first shaft includes a rotation support portion that extends from a keyed end to a bushing mount and a fixed support portion that extends from the bushing mount to an inner end of the first shaft, and wherein the rotational damper is attached to the fixed support portion of the first shaft.

4. The armrest assembly of claim 3, further including a bushing that is attached to the bushing mount and fixed on the first shaft, wherein the bushing is located between the rotation support portion of the first shaft and the armrest body.

5. The armrest assembly of claim 3, wherein the fixed support portion of the first shaft is located in the interior space of the armrest body.

6. A seat assembly comprising: a seat; a first bracket and a second bracket that are each fixed to the seat; an armrest body that defines an interior space; a first shaft that extends from a first side of the armrest body along an armrest axis and is fixed to the first bracket, wherein: the armrest body is attached to the first shaft for relative rotational movement about the armrest axis; a second shaft that extends from a second side of the armrest body along the armrest axis and is fixed to the second bracket, wherein the armrest body is attached to the second shaft for relative rotational movement about the armrest axis; and a rotational damper located in the interior space of the armrest body, the rotational damper including a first damper part and a second damper part attached to the first damper part for relative rotational movement, wherein the first damper part is fixed to the first shaft and the second damper part is fixed to the armrest body.

7. The seat assembly of claim 6, wherein the armrest body is attached to the seat for relative rotational movement from a stowed position through a first range of motion and a second range of motion to a usage position, and wherein the rotational damper is adapted to limit movement of the armrest relative to the seat through the second range of motion.

8. The seat assembly of claim 6, wherein the first shaft includes a keyed end that engages the first bracket to fix the first shaft relative to the first bracket.

9. The seat assembly of claim 6, wherein the first shaft includes a rotation support portion that extends from the first bracket to a bushing mount and a fixed support portion that extends from the bushing mount to an inner end of the first shaft, wherein the rotational damper is attached to the fixed support portion of the first shaft.

10. The seat assembly of claim 9, further including a bushing attached to the bushing mount and fixed on the first shaft, wherein the bushing is located between the rotation support portion of the first shaft and the armrest body.

11. The seat assembly of claim 9, wherein the fixed support portion of the first shaft is located in the interior space of the armrest body.

12. An armrest assembly comprising: an armrest body that defines an interior space; a first shaft that extends from a first side of the armrest body along an armrest axis and is attached to the armrest body for relative rotational movement about the armrest axis, the first shaft including a rotation support portion that extends from a keyed end to a bushing mount, and a fixed support portion that extends from the bushing mount to an inner end of the first shaft, wherein the fixed support portion of the first shaft is located in the interior space of the armrest body; a bushing attached to the bushing mount and fixed on the first shaft, wherein the bushing is located between the rotation support portion of the first shaft and the armrest body; a second shaft that extends from a second side of the armrest body along the armrest axis and is attached to the armrest body for relative rotational movement about the armrest axis; and a rotational damper located in the interior space of the armrest body, the rotational damper including a first damper part and a second damper part attached to the first damper part for relative rotational movement, wherein the first damper part is fixed to the fixed support portion of the first shaft and the second damper part is fixed to the armrest body.

